



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2014

BIOLOGY

Higher 1

Paper 2 Core Paper

8875/02

Thursday, 28 August

2 hours

Candidate Name: _____

Civics Group: _____

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
Section A	
1	/12
2	/ 9
3	/ 9
4	/10
Section B	
5 / 6	/20
TOTAL	/60

Section A

Answer **all** the questions in this section.

For
Examiner's
Use

- 1 Fig. 1.1 shows an electron micrograph of part of a liver cell.

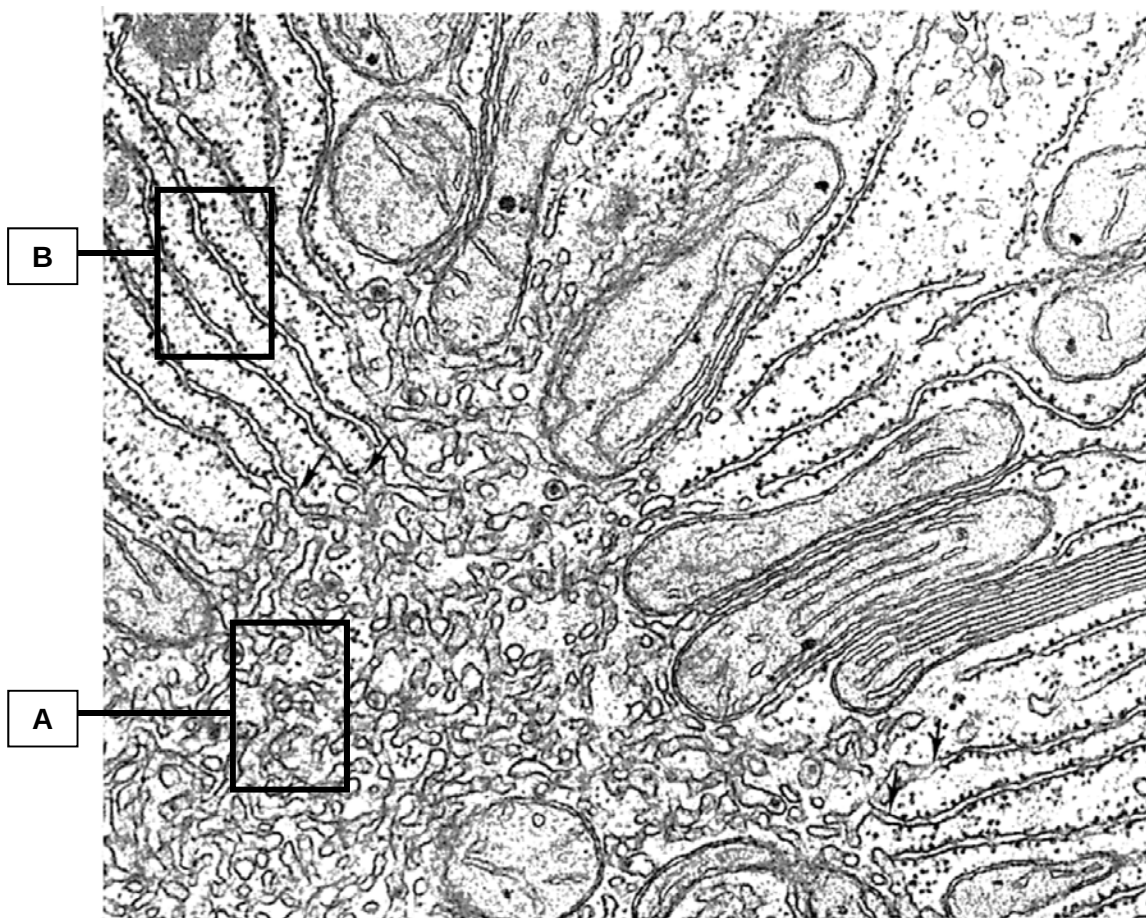


Fig. 1.1

- (a)** Identify structures **A** and **B**.

A: _____

B: _____

[2]

- (b)** State the function of structure **A**.

[1]

[Turn over

(c) Relate the structural features of **B** to its function.

[2]

Cholesterol is taken up by liver cells, which transforms it into 30 to 100 nm low-density lipoprotein (LDL) particles. Fig. 1.2 shows the structure of an LDL particle. It consists of a core of triglycerides and a surface coat of phospholipids, cholesterol and a monomeric protein, ApoB-100.

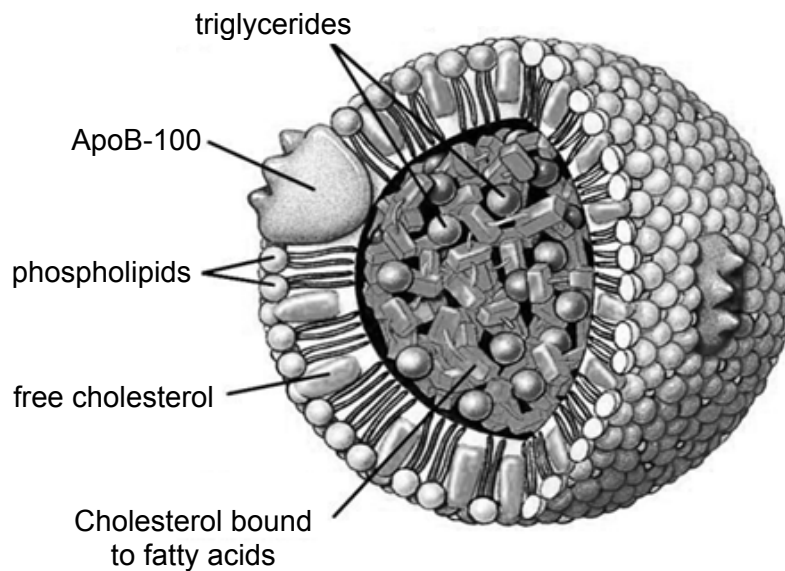


Fig. 1.2

(d) With reference to Fig. 1.2, outline the roles of lipids in the LDL coat.

[3]

(e) Fig. 1.3 shows the three dimensional structure of ApoB-100.

For
Examiner's
Use



Fig. 1.3

(i) Label the type(s) of secondary protein structure observed in Fig. 1.3; [1]

(ii) Explain how the structure of ApoB-100 protein is maintained.

[3]

[Total: 12]

[Turn over

- 2 In DNA replication, primers are added, then progressively removed and replaced by DNA nucleotides for both the leading strand and the lagging strand.

(a) State **one** similarity and **one** difference between the synthesis of the leading strand and the lagging strand during DNA replication.

[2]

(b) Give **one** reason why the primer removed from the 5' end of the daughter strand is **not** replaced by DNA nucleotides.

[1]

The replication of DNA requires enzymes and substrates. One of the substrates is ATP.

However it has been found that 9- β -D-Arabinofuranosyladenine (ara-ATP) is a chemical that affects the activity of an important enzyme in the process.

In an investigation, the effect of different concentrations of ATP on the rate of DNA synthesis was determined:

- with no ara-ATP
- with a low concentration of ara-ATP
- with a high concentration of ara-ATP

The results of the investigation are shown in Fig. 2.2.

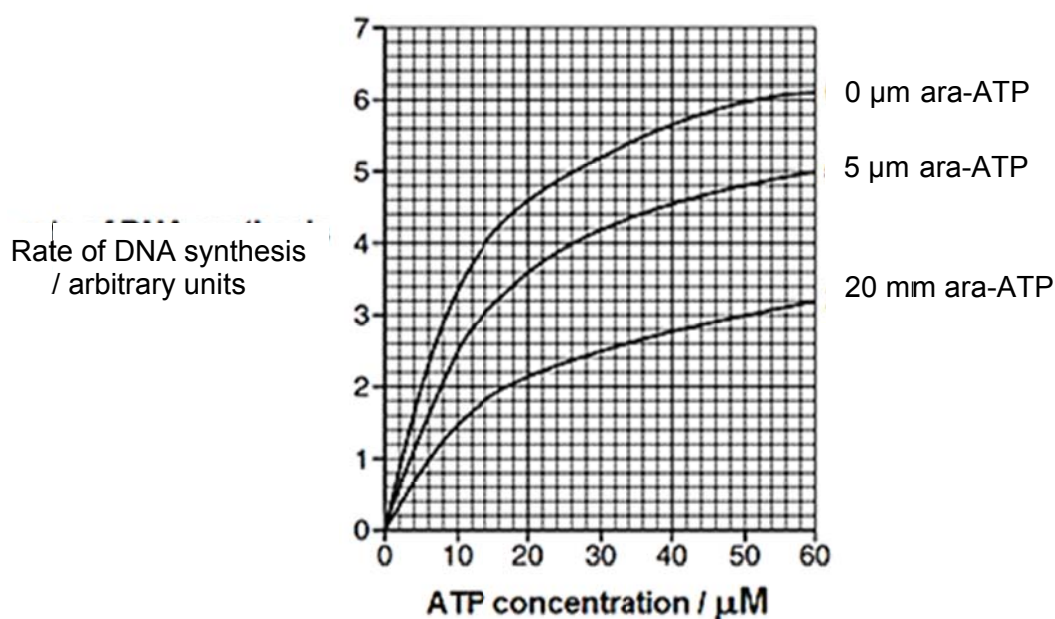


Fig. 2.2

(c) Explain the effects of ara-ATP on the rate of DNA synthesis.

[4]

- (d) State **two** ways in which polymerase chain reaction (PCR) is similar to the replication of DNA.

[2]

[Total: 9]

For
Examiner's
Use

- 3 Fig. 3.1 shows the pedigree chart of a family with congenital alactasia, also known as congenital lactase deficiency. It is a disorder in which infants are unable to break down lactose in milk and can result in severe diarrhoea, dehydration and weight loss.

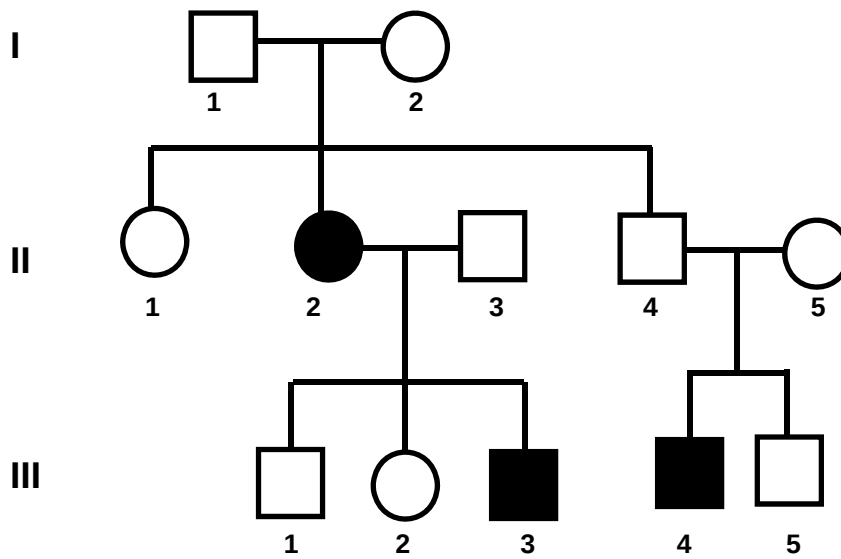


Fig. 3.1

- (a) With reference to Fig. 3.1, explain the mode of inheritance of the allele for congenital alactasia.

[2]

- (b) Using suitable symbols, state the possible genotype(s) of individuals II-2 and III-5.

II-2 : _____

III-5 : _____

[1]

The *LCT* gene codes for lactase. This enzyme helps to digest lactose, a sugar found in milk and other dairy products. At least nine *LCT* gene mutations cause congenital lactase deficiency (congenital alactasia). The *LCT* gene mutations either change the polypeptide sequence of lactase or result in an enzyme that is abnormally short.

(c) Describe how a gene mutation can result in an enzyme that is abnormally short.

[2]

The domestication of plants and animals roughly 10,000 years ago profoundly changed human diets. In Europe, the domestication of cattle brought about an increase in nutrition from cow's milk.

Several thousand years later, a genetic mutation spread through Europe that gave people the ability to produce lactase. This adaptation opened up a rich new source of nutrition that could have sustained communities when harvests of crops failed.

At present, more than 90 % of adults in Britain and Scandinavia have the ability to break down lactose found in milk.

(d) Explain how the process of natural selection could have resulted in a large proportion of adults in Europe with the ability to break down lactose.

[4]

[Total: 9]

- 4 Photosynthetic and metabolic acclimation to low temperatures were studied in *Arabidopsis*. A group of plants were grown at 23 °C and then shifted to 5 °C (i.e. 23/5 °C). The leaves were collected and frozen at the temperature of liquid nitrogen. Enzyme activity in the leaves was then compared to those grown in 23 °C and in 5 °C.

The activity of the C₃-cycle enzymes (Rubisco, PGK and GAPDH) is presented in Fig. 4.1.

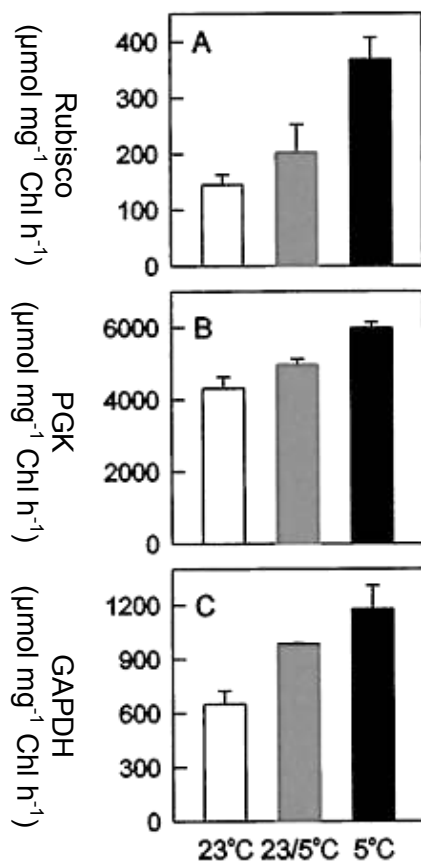


Fig. 4.1

- (a) Compare the effect of temperature on the activity of the C₃-cycle enzymes.

[2]

- (b) State the role of Rubisco in the C₃-cycle.

[1]

(c) Explain the effect of high oxygen concentration on Rubisco and on photosynthesis.

[2]

(d) Suggest why the leaves had to be frozen at the temperature of liquid nitrogen.

[1]

In the same research, the cells from the different plants were studied using electron micrographs (shown in Fig. 4.2).

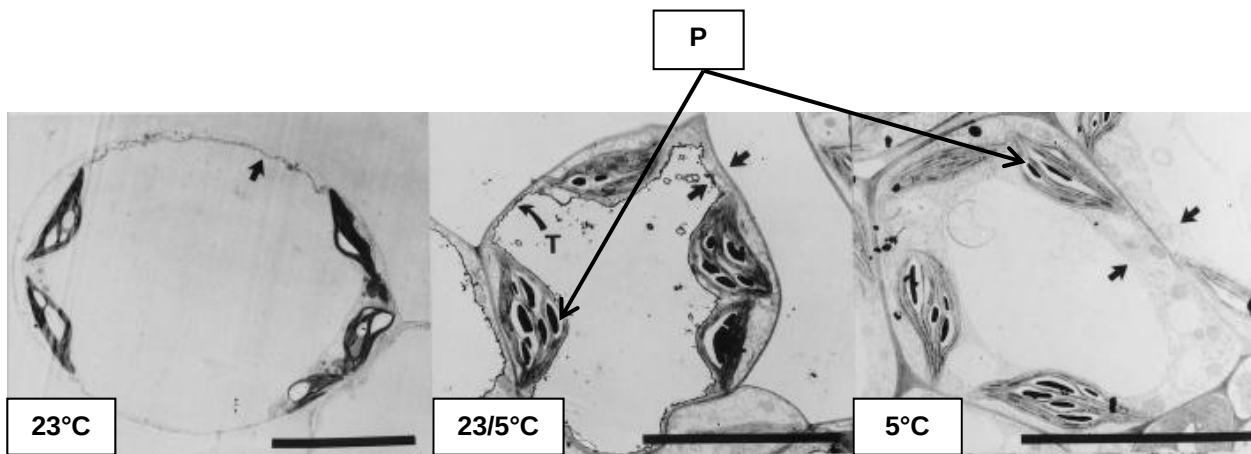


Fig. 4.2

(e) Identify structure P.

[1]

- (f) Describe the effect of temperature on the relative volume of cytoplasm and the relative volume of vacuole.

For
Examiner's
Use

[3]

[Total: 10]

END OF SECTION A

[Turn over

Section B: Essay Questions

There are 2 questions in this section. Answer **only one** question.

Write your answer on the separate answer paper provided. At the end of the examination, fasten the separate answer paper securely.

You are advised to spend not more than 30 minutes on this section.

Your answer should be illustrated by large, clearly labelled diagrams wherever suitable.

- 5 (a) Describe how cellulose differs from starch. [6]
- (b) Temperature is a factor that affects the permeability of living cells. Using a beetroot that has red pigments in its cells, describe an investigation into the effect of temperature on the permeability of the cell. [9]
- (c) Explain how uncontrolled cell division can result in cancer. [5]

[Total: 20]

- 6 (a) Describe the properties of plasmids that allow them to be used for DNA cloning. [7]
- (b) Discuss the social and ethical issues of the Human Genome Project. [8]
- (c) Describe the features of blood stem cells and explain their normal functions. [5]

[Total: 20]

END OF SECTION B

BLANK PAGE